

SIMPLE LINEAR REGRESSION PROBLEM.

Answer.

Given dataset,

load (N)	Extension (mm)
5	1.2
10	2.5
15	3.7
20	4.9
25	6.1

let load (N) = x and Extension (mm) = y ,

x	y
5	1.2
10	2.5
15	3.7
20	4.9
25	6.1

We know for Simple linear Regression, we use the formula $y = mx + c$, where m is slope and c is y -intercept,

The equation for m and c is,

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum (x^2) - (\sum x)^2}$$

$$c = \frac{\sum y - m \sum x}{n}$$

So we have to find $\sum xy$, $\sum x \sum y$, $\sum (x^2)$, $\sum y$, $\sum x$ in order to find m and c .

x	y	xy	x^2
5	1.2	6	25
10	2.5	25	100
15	3.7	55.5	225
20	4.9	98	400
25	6.1	152.5	625

Then,

$$\sum x = 75$$

$$\sum y = 18.4$$

$$\sum x \sum y = 1380$$

$$\sum xy = 337$$

$$\sum x^2 = 1375$$

We know, $n = 5$

$$\therefore m = \frac{n \sum xy - \sum x \sum y}{n \sum (x^2) - (\sum x)^2}$$

$$= \frac{5 \times 337 - 1380}{5 \times 1375 - (75)^2}$$

$$= \frac{5 \times 337 - 1380}{5 \times 1375 - 5625}$$

$$= \frac{1685 - 1380}{6875 - 5625}$$

$$= \frac{305}{1250}$$

$$m = 0.244 \quad \text{---(1)}$$

$$\text{and } c = \frac{\sum y - m \sum x}{n}$$

$$= \frac{18.4 - 0.244 \times 75}{5} = \frac{18.4 - 18.3}{5}$$

$$c = \frac{0.1}{5} = 0.02 \quad \text{---(2)}$$

Coming back to the equation of line,

$$y = mx + c$$

Substituting (1) and (2) in the equation,

$$y = (0.244)x + 0.02$$

→ Simple regression
ML model

where $x = \text{load (N)}$

and $y = \text{Extension (mm)}$

Now to predict the extension when applied load is 55 N,

we know $x = 55 \text{ N}$ so we need to find y .
For that we use that in the equation

$$y = (0.244)x + 0.02$$

$$y = (0.244)(55) + 0.02$$

$$= 13.42 + 0.02$$

$$y = 13.44$$

$y = 13.44 \text{ mm}$ is the predicted Extension at 55 N